Named Entity Recognition in German-language Telegram channels

Bachelor Thesis

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by

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Abstract

This thesis implements Named Entity Recognition (NER) in German-language text data dissemninated via the messenger service Telegram, with a focus on the mobilization for protests against measures to counteract the COVID-19 in Germany.

This work explores the potentials and limitations of two existing German-language models for NER, namely *spacy* and *hugging face*, for the recognition of the entity types PERSON, LOCATION, and ORGANIZATION.

Furthermore, it makes use of the ctparse module to extract text parts containing DATE and TIME information.

This thesis contributes to the task of German-language NER in general by applying it to a specific domain and data type, namely text data from messenger services. Furthermore, its aim is to provide a low-resource tool for both researchers and civil society for monitoring and analyzing large amount of text data from social media/messengers.

Dedication

Statutory Declaration

I herewith formally declare that I have written the submitted thesis independently. I did not use any outside support except for the quoted literature and other sources mentioned in the paper.
I clearly marked and separately listed all of the literature and all of the other sources which I employed when producing this academic work, either literally or in content.
I am aware that the violation of this regulation will lead to the failure of the thesis.

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Contents

Chapter 1

Introduction

1.1 Background

The outbreak of the COVID-19 pandemic was followed by an increasing spread of misinformation and conspiracy theories. Those narratives evolved not only around the virus itself, but also about the intentions of those responsible for putting into effect measures for its containment.

Those measures included the mandatory use of face masks, rapid testing, and vaccinations, at times functioning as admission requirements in the workplace or publicly accessible spaces such as restaurants and theatres. Furthermore, unprecedented restrictions were imposed on public life through several general or partial lock-downs of social and economic life, as well as temporary bans of gathering in public. These restrictions also heavily affected the possibilities to hold political demonstrations.

From the beginning on, these measures were accompanied by critique and protests from various political spectres. In Germany, the first noticeable protests against COVID-19 measures took place at the so-called 'Hygienedemonstrationen' (hygiene demonstrations) in front of the 'Volksbühne' theatre in Berlin. Over time, the so-called 'Querdenken' (lateral thinking) movement became the most prominent movement evolving around protests against the COVID-19 containment measures. The movement reached its preliminary climax with several large demonstrations taking place in Berlin, directed against legislative changes regarding the containment of the pandemic passed by the German parliament. In the context of these demonstrations, violent clashes occurred between police and protesters, which often refused to act according the official restrictions for the demonstration such as wearing a face mask. Furthermore, there were several incidents in which journalists were violently attacked by protesters, accusing them to be part of the 'Lügenpresse' (lying press), and two incidents in which the seat of the German parliament, the Reichstag, was targeted by protesters: On August 29th 2020, around 500 protesters, among them several prominent actors from the German far-right movement, broke the barriers in front of the Reichstag in Berlin, seat of the German parliament, and occupied the stairs of the building, waving several 'Reichskriegsflaggen'. On November 18th 2020, another large demonstration organized by the 'Querdenken' movement took place; once again accompanied by heavy clashes between protesters and police officers. On the same day, a small group of right-wing social media activists was invited as visitors to the Reichstag by the German far-right party 'Alternative für Deutschland' (AfD). The group approached and harassed politicians from other parties, filming and live-streaming the events, an incident which was perceived as a caesura and heavily criticized by the other parties.

On April 21st 2021, the last large authorized demonstration of the 'Querdenken' movement took place in Berlin, comparing a new law for the containment of the pandemic with the 'Ermächtigungsgesetz' put into effect in 1933 which ultimately paved the way for the Nazi dictatorship in Germany. In response to several violations of official restrictions by demonstration participants, police decided to disperse the demonstration. Several protesters subsequently moved to other areas nearby. Clashes between protesters and police continued to occur throughout the rest of the day.

The subsequent demonstrations announced by the 'Querdenken' movement to take place in Berlin were prohibited by authorities, arguing with the experience at prior demonstrations that participants would act against against the official restrictions. The ban was confirmed by Berlin's administrative court after the organizers of the protest had filed a suit against the initial prohibition. In spite of the ban, several hundred protesters gathered in different spots in Berlin e.g. on the 21st, 28th and 29th of August 2021.

The mentioned bans of large demonstrations of the 'Querdenken' movement as well as the general restrictions for demonstrations and public gatherings, limiting the number of participants to a maximum of twenty persons, decreased the momentum of the large, centralized protests focussing on the German legislative located in Berlin. However, the protests did not stop as a result, but rather changed their character: They became more decentralized and a shift to more informal forms of gatherings and protests could be observed: In several German cities and towns, people gathered for 'Mahnwachen' or 'Spaziergänge' (walks) to protest against the government's COVID-19 politics. Since going for a walk individually was a legal outdoor activity even under the strictest lock-down regulations, this type of action was chosen in order to circumvent the restrictions imposed on demonstrations. In effect, these walks ('Spaziergänge') often resulted in large gatherings of people moving through the streets together, often holding signs or chanting slogans against German politicians or the COVID-19 containment measures in general.

However, the calls to gather for walks in the public became more targeted and mobilized people to gather in front of the private homes of politicians, mostly those in charge for the COVID-19 measures on federal state level.

In December 2021, around thirty members of the far-right group 'Freie Sachsen' (Free Saxonians) gathered in front of the home of Saxony's Minister of Health, Petra Köpping, carrying torchlights. The incident received a lot of public attention, and was criticized in the media as well as by German politicians as attempts to intimidate politicians under the pretext of the protesters right to ‘freedom of speech’.

Earlier that year, the same group had gathered in front of Saxony’s prime minister, Michael Kretschmer, who is one of the most prominent targets of the radical movement against the COVID-19 containment measures. In December 2021, an investigative report by German journalists revealed that members of the Telegram chat group ‘Dresden Offlinevernetzung’ had talked about concrete preparations to assassinate Kretschmer. A few months later, by mid-April 2022, police foiled a plot to kidnap Germany’s Minister of Health Karl Lauterbach and to sabotage power facilities in order to create chaos which would ultimately allow to overthrow the government.

The messenger service Telegram played a key role in the mobilization of the movement. The service allows the creation of large groups and channels to disseminate information quickly to a large number of users. Unlike Facebook or Google, which hand over data to authorities at their request, Telegram does not cooperate with authorities.

This ensures a high degree of anonymity for its users which has proven beneficial for protest movements in authoritarian countries, This anonymity poses a barrier to political repression and criminal prosecution. This makes the service attractive not only for social movements, but also for criminal and terrorist activities, and also facilitates the spread of hate-speech and calls for violence like those mentioned above.

The rise of conspiracy theories in the context of the COVID-19 pandemic has thus not only intensified hate-speech against and the discrimination of certain communities in the form of antisemitism and racism - this process was accompanied by an increasingly toxic and potentially violent climate for politicians in Germany. This makes it once more clear that the digital sphere of social networks and messengers cannot be grasped without also considering its entanglement with and effects for the physical sphere of society. Both online and offline contexts influence and often reinforce each other, as is the case with the mobilization against the COVID-19 containment measures.

The processes illustrated above also point out another important aspect: Even though political activists from Germany's extremist far-right had been involved in the protests from the start, and non-governmental organizations had warned about the antidemocratic and violent potential of parts of the movement, the dangerous potential of this movement seems to have been underestimated by German authorities for a rather long time. Only after journalists had revealed the plans to murder Michael Kretschmer, and after several gatherings in front of private homes of politicians had already taken place, the issue seems to become more of a serious concern.

Thus the process points out the relevance of investigative work by critical journalists and non-governmental organizations to monitor the mobilization and networking strategies of the far-right spectre in Germany. However, while these actors often have to deal with large amount of data, their investigation often relies on manually conducted research and data evaluation. The potentials of Data Science methods, especially Machine Learning-based approaches for Natural Language Processing (NLP), are often not accessible for these actors, making it hard for them to process large amounts of data from the often very dynamic contexts of informal political networks.

- racism: especially Asian community made responsible for spreading the virus
- antisemitism: Jews identified as culprits behind the pandemic, using it as a means to obtain world dominance
- politicians: complicit with those who benefit from pandemic; following a secret plan; restriction of civil rights
- freedom of speech, criticizing the government are valuable and essential components of a democratic society
- but: often hateful tendencies and developments, both online and offline

- entanglement of digital and real world
- hate does not stay in the internet, also has real effects
- spread of misinformation, conspiracy theories: social media and messengers
- not only information, also mobilization

1.2 Objective

This thesis aims at exploring the potentials of a particular NLP technology, namely Named Entity Recognition (NER) for the investigation of the mobilization for protests against COVID-19 measures in a German-language Telegram channel. The overarching goal is to explore if a component for Named Entity Recognition could be a beneficial contribution to the Open Source project 'Teledash', to make NER available as a tool for journalists, NGOs, and researchers investigating the COVID-19 related protest movements and networks in Germany.

- low-resource approach of NER in German-language Telegram messages
- extraction of entities particularly relevant for mobilization / monitoring of potential targeting: PERSON, ORGANIZATION; DATE; TIME; LOCATION; ACTION as custom type
- explore potentials and limitations of existing models
- explore potentials of translation approach
- provide a low-resource tool for NGOs and researchersm make available as API; Project teledash

1.3 Thesis Structure

- Theory
- Implementation
- Conclusion and Discussion

Chapter 2

Theory

The first part of this chapter places Named Entity Recognition in its larger context of Information Extraction (IE) technology, and provides a basic definition of of NER. Furthermore it briefly describes the historical evolvement of NER, and discusses the challenges related to this task, both in general and with regard to the peculiarities of German language.

The second part elaborates different approaches for NER, from rule-based methods over statistical machine-learning based approaches to neural networks including transformer models, and discusses their respective limitations and potentials.

2.1 Information Extraction

Named Entity Recognition is a task from the field of Information Extraction (IE). IE aims at extracting factual information from unstructured or semistructured natural language texts. (cf. Zong et al. 2021, 227)

The extracted parts of information can be entities, attributes related to those entities, relationships between entities, or events. The kinds of texts information is extracted from may be news texts from the web, social media posts, customer reviews, academic literature, and others. The aim of IE is to generate structured output from this unstructured or semistructured text data. Typical tasks of IE are named entity recognition, entity disambiguation, relation extraction, and event extraction (Zong et al. 2021)

The origin of IE dates back to the late 1970s.

- MUC - CoNLL

Common applications of information extraction are for example news tracking, counter-terrorism, business and financial intelligence, the processing of biomedical data or scientific libraries. IE can also be utilized for applications building on entity-oriented search, for question-answering systems, or the task of text segmentation. (Aggarwal 2018)

IE technology can be classified via the domain aspect (limited vs. open domain), the type of extracted information (entity recognition, relation extraction, event extraction), and implementation methods (rule-based, statistical learning-based, or deep learning-based). (Zong et al. 2021, 229)

2.2 Named Entity Recognition

2.2.1 Definition

Named entity recognition is the task of identifying named entities like person, location, organization, drug, time, clinical procedure, biological protein, etc. in text. It is often used as the first step in question answering, information retrieval, co-reference resolution, topic modeling, etc (Yadav and Bethard 2019) and is a "key component in NLP systems for question answering, information retrieval, relation extraction, etc." (Yadav and Bethard 2019)

NER is a fundamental task in natural language processing. It aims at identifying entities belonging to specified categories in text. The respective entities can mainly be represented by the following categories:

- PERSON

- LOCATION
- ORGANIZATION
- TIME
- DATE
- CURRENCY/QUANTITY number, PERCENTAGE

While time, date, currency/quantity number, and percentage are commonly recognized via regular expressions, person, location, and organization pose larger challenges. Thus most of NER research focusses mainly on these entity types. (Zong et al. 2021, 229)

NER can be divided into two subtasks:

- entity detection = detect whether a word string in a given text is an entity
- entitiy classification = judge the specified category of the detected entities (Zong et al. 2021, 229)

Aggarwal 2018

2.2.2 Historical evolvement

first NER task: by Grishman and Sundheim (1996) in the Sixth Message Understanding Conference since then numerous NER tasks ((Tjong Kim Sang and De Meulder, 2003; Tjong Kim Sang, 2002; Piskorski et al., 2017; Segura Bedmar et al., 2013; Bossy et al., 2013; Uzuner et al., 2011) (Yadav and Bethard 2019)

2.2.3 Challenges for NER

- ambiguity - abbreviations - new entities evolving - capitalization - social media: messy texts, typos, codes - mixed language texts - domain adaptation

2.3 Overview of NER approaches

Early NER systems were based on handcrafted rules, lexicons, orthographic features and ontologies. followed by NER systems based on feature-engineering and machine learning (Nadeau and Sekine, 2007) Starting with Collobert et al. (2011), neural network NER systems with minimal feature engineering have become popular are appealing because they typically do not require domain specific resources like lexicons or ontologies, and are thus poised to be more domain independent Various neural architectures have been proposed, mostly based on some form of recurrent neural networks (RNN) over characters, sub-words and/or word embeddings. (Yadav and Bethard 2019)

2.3.1 Rule-based approaches

Each token in the text is converted into a set of features. These features are typically helpful properties of the token or their context for entity extraction. e.g. starts with capital letter help in defining various patterns on the left-hand side of the rule contextual pattern on the left-hand side of the rule is a combination of conditions corresponding to the features associated with a sequence of tokens a rule is fired if a sequence of tokens in the text matches this pattern action on the right-hand side could correspond to labeling that sequence as a named entity (Aggarwal 2018)

2.3.2 Knowledge-based approaches

Knowledge-based NER systems do not require annotated training data as they rely on lexicon resources and domain specific knowledge. These work well when the lexicon is exhaustive, but fail for entity classes not represented in the lexicon Precision is generally high for knowledge-based NER systems because of the lexicons, but recall is often low due to domain and language-specific rules and incomplete dictionaries Another drawback of knowledge based NER systems is the need of domain experts for constructing and maintaining the knowledge resources. (Yadav and Bethard 2019)

2.3.3 Machine-learning based

Feature-engineered supervised systems Supervised machine learning models learn to make predictions by training on example inputs and their expected outputs, and can be used to replace human curated rules. Common ML systems for NER: Hidden Markov Models (HMM) Maximum Entropy Support Vector Machines (SVM) Conditional Random Fields (CRF) decision trees (Yadav and Bethard 2019)

2.3.4 Neural Networks

NER systems have been studied and developed widely for decades but accurate systems using deep neural networks (NN) have only been introduced in the last few years (Yadav and Bethard 2019)

Feature-inferring neural network systems Collobert and Weston (2008) proposed one of the first neural network architectures for NER, with feature vectors constructed from orthographic features (e.g., capitalization of the first character), dictionaries and lexicons. Later work replaced these manually constructed feature vectors with word embeddings (Collobert et al., 2011), which are representations of words in n-dimensional space, typically learned over large collections of unlabeled data through an unsupervised process such as the skip-gram model (Mikolov et al., 2013). Studies have shown the importance of such pre-trained word embeddings for neural network based NER systems (Habibi et al., 2017), and similarly for pre-trained character embeddings in character-based languages like Chinese Modern neural architectures for NER can be broadly classified into categories depending upon their representation of the words in a sentence. For example, representations may be based on words, characters, other sub-word units or any combination of these (Yadav and Bethard 2019)

2.3.5 NER Datasets

NER datasets first shared task on NER (Grishman and Sundheim, 1996) CoNLL 2002 (Tjong Kim Sang, 2002) and CoNLL 2003 (Tjong Kim Sang and De Meulder, 2003) were created from newswire articles in four different languages (Spanish, Dutch, English, and German) and focused on 4 entities - PER (person), LOC (location), ORG (organization) and MISC (miscellaneous including all other types of entities). NER shared tasks have also been organized for a variety of other languages; German (Benikova et al., 2014) named entity types vary widely by source of dataset and language Benikova et al. (2014)'s data, which is based on German wikipedia and online news, has named entity types similar to that of CoNLL 2002 and 2003: PERson, ORGanization, LOCation and OTHer NER tasks have also been organized on social media data, e.g., Twitter, where the performance of classic NER systems degrades due to issues like variability in orthography and presence of grammatically incomplete sentences (Baldwin et al., 2015) Entity types on Twitter are also more variable (person, company, facility, band, sportsteam, movie, TV show, etc.) as they are based on user behavior on Twitter Though most named entity annotations are flat, some datasets include more complex structures; e.g. nested entities (Yadav and Bethard 2019)

2.3.6 Tagging scheme and entity types

B-I-O tagging scheme

2.3.7 Evaluation metrics

precision recall F1

2.3.8 Summarization

Chapter 3

Implementation

3.1 Data

3.1.1 Existing German-language datasets for NER

3.1.2 Domain-specific data

Telegram channel ‘Demotermine’: 52.900 subscribers by the beginning of June 2022,

long channel name: ‘Demo-Termine Kontakte, Gleichgesinnte treffen - Demokalender, Spaziergänge, Mahnwachen, Meditation *Rücktritt Bundesregierung!’

channel-info: “ Wir Demokraten wollen miteinander in Kontakt bleiben @DemotermineChat und uns im Bedarfsfall möglichst schnell austauschen, Beweise in Wort, Bild und Ton sichern, damit sie nicht unterschlagen werden können @LivestreamsFuerDich! @Terminverteiler$_{B}ot'$

TODO insert histplot to show development of message numbers

date range: 2018-10-05 bis 2021-12-19

total number of messages (wo duplicates): 27.627 messages

German-language: 23.911

Sample size: xx out of 23.911 confidence level 95

text length: min 100 (lower quartile), max 350 (upper quartile)
-¿ remaining:

3.2 Models

- spacy - hugging face

3.3 Pipeline

3.4 Evaluation metrics

3.4.1 Accuracy

3.4.2 Performance

3.5 Translation approach?

3.6 Summarization

Chapter 4

Results

4.1 Spacy

4.2 Hugging Face

4.3 Translation approach

4.4 Summarization

Chapter 5

Conclusion

5.1 Summarization

5.1.1 subsection

5.2 Discussion

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Appendix

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